Biophilic VR Project —Participant Questionnaires + Heart-Rate Data Table

Study context

This study evaluates how a short VR biophilic experience (e.g., Arctic map vs MYFYP/Garden map) impacts stress and affect. Participants complete the questionnaires immediately after each condition (or once post-exposure, depending on your design).

Section A. Perceived Stress Scale (PSS-10) —adapted to your VR session

Timeframe: **During the last 6 minutes (this VR session)**, how often did you feel or think the following?

0 = Never 1 = Almost never 2 = Sometimes 3 = Fairly often 4 = Very often

PSS-10 Items

1. In this VR session, how often have you been upset because of something that happened unexpectedly?
2. In this VR session, how often have you felt that you were unable to control important things?
3. In this VR session, how often have you felt nervous or "stressed"?
4. In this VR session, how often have you felt confident about your ability to handle personal problems?
5. In this VR session, how often have you felt that things were going your way?
6. In this VR session, how often have you found that you could not cope with all the things you had to do?
7. In this VR session, how often have you been able to control irritations in your life?
8. In this VR session, how often have you felt that you were on top of things?
9. In this VR session, how often have you been angered because of things outside your control?
10. In this VR session, how often have you felt difficulties were piling up so high that you could not overcome them?

Scoring (for analysis)

Reverse-score items **4, 5, 7, 8** using: 0, 1, 2, 3, 4.

Total PSS-10 score = sum of all 10 items after reversing (range 0–). Higher scores indicate greater perceived stress during the VR session.

Section B. PANAS-style Affect Questionnaire (project-tailored; not verbatim PANAS)

Purpose: measures momentary positive and negative affect after the VR experience, aligned to biophilic/relaxation outcomes.

Timeframe: **Right now (immediately after this VR session)**.

Response scale (circle one):

1 = Very slightly / Not at all 2 = A little 3 = Moderately 4 = Quite a bit 5 = Extremely

Positive Affect (PA)

PA1. Engaged

PA2. Interested

PA3. Energised

PA4. Enthusiastic

PA5. Inspired

PA6. Calm

PA7. Content

PA8. Refreshed

PA9. Focused

PA10. Comfortable

Negative Affect (NA)

NA1. Tense

NA2. Irritable

NA3. Overwhelmed

NA4. Uneasy

NA5. Frustrated

NA6. Worried

NA7. Restless

NA8. Disappointed

NA9. Mentally fatigued

NA10. Stressed

Scoring (for analysis)

PA score = sum (or mean) of PA1.

NA score = sum (or mean) of NA1.

Note: This section is **PANAS-style and** suitable for academic evaluation, but it does not reproduce the copyrighted PANAS item set verbatim.

Section C. Optional VR Experience Checks

Presence / immersion (short)

P1. I felt like I was “really there”in the virtual environment. (1–5)

P2. The virtual environment felt believable. (1–5)

P3. I paid attention to the environment rather than the real room. (1–5)

Cybersickness (short)

C1. I felt nauseous. (1–5)

C2. I felt dizzy. (1–5)

C3. I experienced eyestrain / headache. (1–5)

Sound audibility

A1. I could clearly hear environmental audio in this map. (1–5)

Section D. Heart Rate Monitor Data Table

Use one row per participant per condition. If you have continuous data, store the raw time-series separately and summarise here.

Table D1: Summary metrics

ID	Start time	Duration (min)	Baseline HR (bpm)	Mean HR (bpm)	Peak HR (bpm)